

Exponent and multiplicity are robust for every bounded ξ :

$$Z(a + J) = \Theta(n^{-p/2}(\log n)^m)$$

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Abstract

For an arbitrary continuous J on the box, with no divisibility assumption, the standard integral with $\xi = a + J$ is sandwiched between the constant- ξ integrals at $a \pm L$, $L = \sup |J|$, because $e^{\beta t \xi}$ is monotone in ξ for $t \geq 0$. Both bounds are $\sim C_{\pm} n^{-p/2}(\log n)^m$ by unit 59, so $Z(a + J) = \Theta(n^{-p/2}(\log n)^m)$: the exponent and the logarithmic degree of `thm:TaylorTree` are determined by (h, k) alone for every bounded ξ , and only the constant depends on ξ . Zero `sorry/axiom`.

1 The things formalised

```
theorem boxIntegralXi_sandwich ... (hJL : u, ( i, 0 u i u i b) → |J u| L) :
  boxIntegralFin (a - L) b c k h boxIntegralXi a b c k h J
  boxIntegralXi a b c k h J boxIntegralFin (a + L) b c k h
theorem boxIntegralXi_isTheta ( a b : ) (h : 0 < ) (hb : 0 < b) (D : )
(k h : Fin (D + 2) → ) (hk : i, 0 < k i) (J : (Fin (D + 2) → ) → )
(hJ : Continuous J) :
  (p : ) (m : ), ( i, p finExp k h i)
  m + 1 = (Finset.univ.filter fun i => finExp k h i = p).card
  (fun n => boxIntegralXi a b (Real.sqrt n) k h J)
  =θ[atTop] fun n => n ^ (-(p / 2)) * Real.log n ^ m
```

2 Proof strategy

Pointwise monotonicity of the exponential, integrated with `integral_mono_ae` on the box; then `IsBig0.antisymm` from $Z(\xi) = O(Z(a+L)) = O(g)$ and $g = O(Z(a-L)) = O(Z(\xi))$, the constant- ξ equivalences supplying the two O -relations with g .

3 Roadblocks and resolutions

- Subscript $\pm$ signs are not identifier characters in Lean 4.
- `isBig0_self_const_mul` takes the nonvanishing proof first.

4 Outcome

Together with units 59, 65 and 66: the leading exponent and multiplicity are universal in ξ and $\eta > 0$ (as Θ -statements), and are sharp asymptotic equivalences with explicit constants for constant ξ and for perturbations divisible by u^k .